

Networks and Systems Management



Network Function Virtualization (NFV)

Filipe Sousa

FEUP – GRS/M.EIC

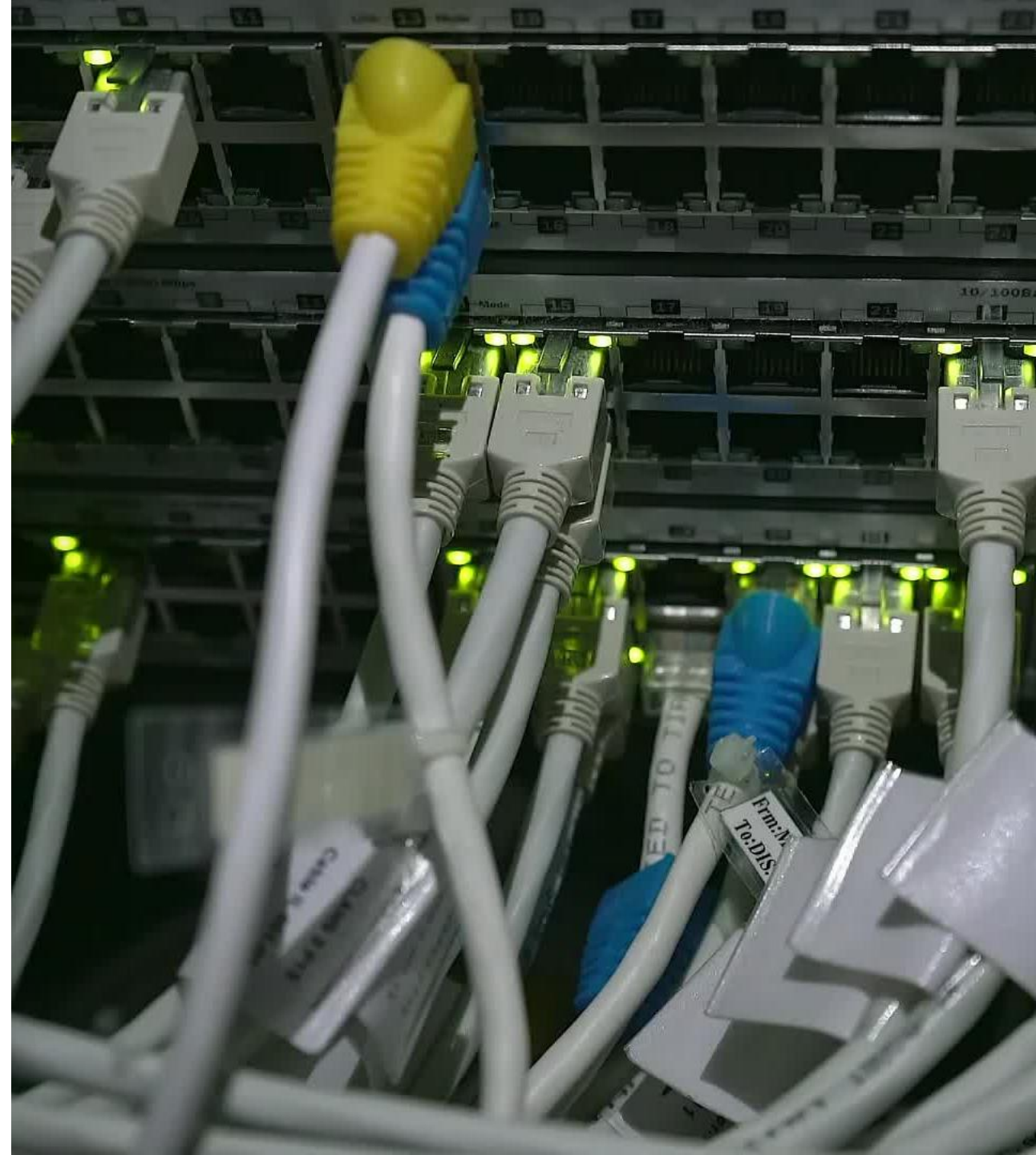
Introduction

- Evolution of networking: from hardware appliances to virtualized services
- Need for flexible, scalable, and cost-effective network management
- Introduction to two key paradigms:
 - Network Function Virtualization (NFV)
 - Software Defined Networking (SDN)



Traditional Network Architecture

- Hardware-dependent appliances (firewalls, routers, load balancers)
- Proprietary systems with tight coupling between hardware and software
- Challenges:
 - Expensive to scale
 - Vendor lock-in
 - Difficult to automate or program



What is NFV?

- Virtualization of network functions (e.g., firewall, NAT, DPI)
- Decouples network functions from proprietary hardware
- Runs on standard x86 servers in data centers or at the edge
- Enables:
 - Faster deployment
 - Resource optimization
 - Dynamic scaling



NFV Architecture

- Key components:
 - VNF (Virtualized Network Function): Software implementation of a network function
 - NFVI (NFV Infrastructure): Hardware and virtualization layer
 - MANO (Management and Orchestration): Controls VNFs and infrastructure
- Based on ETSI NFV framework



What is SDN?



Separation of control plane and data plane



Centralized control via an SDN Controller



Network devices become simple packet forwarding elements



Enables:

- Centralized network visibility
- Policy-driven traffic management
- Rapid innovation and automation

SDN Architecture

Data Plane: Network devices
(switches, routers)

Control Plane: SDN Controller (e.g.,
OpenDaylight, ONOS)

Application Plane: Network apps
(e.g., routing, QoS, security)

Communication via standard APIs
(e.g., OpenFlow)

NFV vs SDN

Feature	NFV	SDN
Purpose	Virtualize network functions	Separate control and data plane
Focus	Network services	Network control
Components	VNFs, NFVI, MANO	SDN Controller, switches
Standards Body	ETSI	ONF (Open Networking Foundation)
Flexibility	High (service chaining, scaling)	High (centralized control)
Common Use Cases	vFirewall, vRouter, vLoadBalancer	Traffic engineering, QoS

NFV and SDN – Complementary Technologies



NFV provides virtualized services



SDN provides programmable control of network traffic



Together, they enable:

- Automated service provisioning
- Network slicing
- End-to-end service orchestration

Use Cases of NFV and SDN



Telecom Networks (5G Core, vRAN)



Enterprise WAN optimization



Cloud-native data centers



Security (DDoS mitigation, Intrusion Detection)



IoT and Edge Computing deployments

An abstract digital cityscape rendered in shades of blue. The scene is composed of numerous rectangular blocks and cubes of varying sizes, some of which are illuminated from within, creating a glowing effect. The surfaces of these blocks are covered in a dense pattern of binary code (0s and 1s). Several bright, colorful beams of light (red, green, and blue) emanate from different points within the structure, suggesting data flow or network activity. The overall composition is dynamic and futuristic, representing a digital or network environment.

Industry Examples

- AT&T: Network virtualization with OpenNFV
- Verizon: Using NFV and SDN for 5G and IoT
- Google B4 Network: Global SDN WAN
- Nokia & Ericsson: NFV-based telecom infrastructure

Challenges and Limitations

- NFV:
 - Performance overhead from virtualization
 - Interoperability between VNFs
 - Complex orchestration
- SDN:
 - Scalability and reliability of centralized control
 - Security concerns (controller attacks)
 - Integration with legacy infrastructure



Key Takeaways

NFV and SDN are transformative technologies for modern networks

NFV focuses on **virtualizing services**, SDN on **programmable control**

Both aim to make networks more agile, cost-effective, and scalable

Used together, they enable **end-to-end network automation**

Further Reading & References

- ETSI NFV White Papers:
<https://www.etsi.org/technologies/nfv>
- ONF SDN Architecture: <https://opennetworking.org>
- "Network Function Virtualization: Challenges and Opportunities" – IEEE Communications
- "Software-Defined Networking: A Comprehensive Survey" – ACM Computing Surveys
- O'Reilly Book: "SDN: Software Defined Networks" by Thomas D. Nadeau
- Cisco White Papers on NFV & SDN
<https://www.cisco.com/c/en/us/solutions/collateral/service-provider/network-functions-virtualization-nfv-infrastructure/white-paper-c11-736645.html>
- IEEE Communications Magazine (NFV/SDN special issues) <https://sdn.ieee.org/publications?showall=1>

